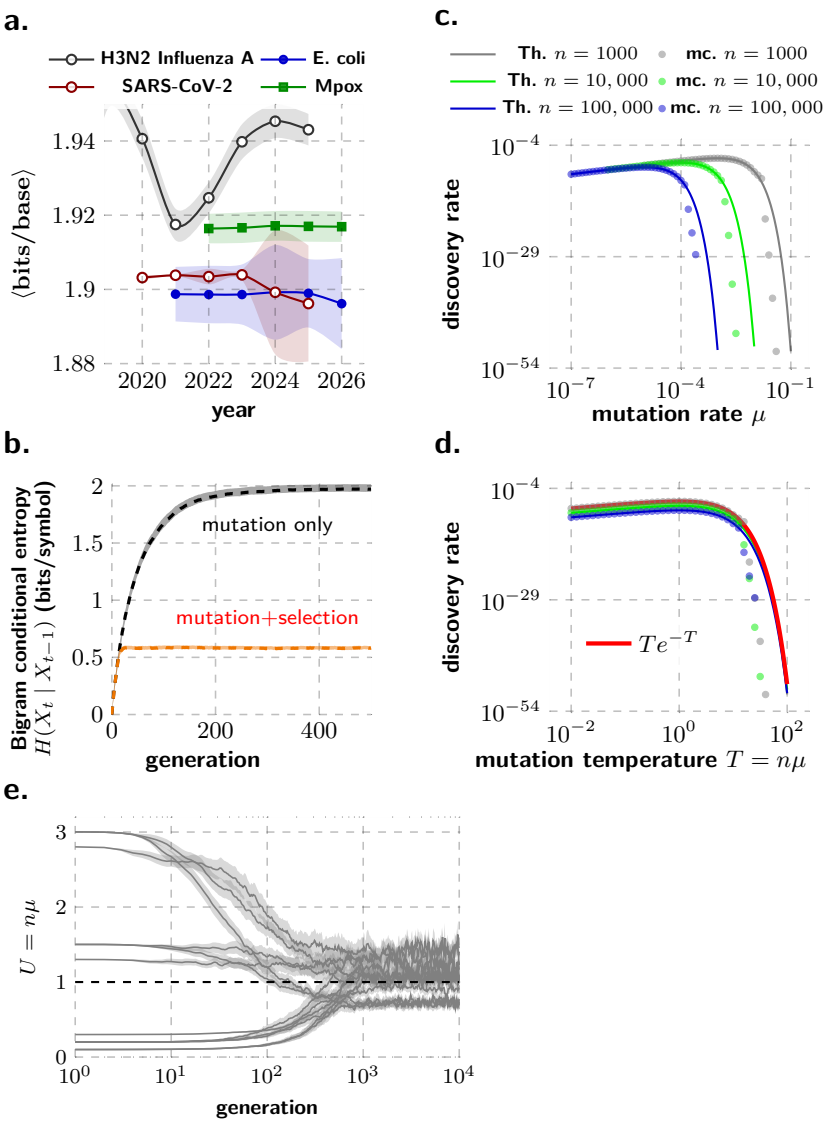
a.
H3N2 Influenza A
E. coli
SARS-CoV-2
Mpox
⟨bits/base⟩
1.94
1.92
1.9
1.88
2020
2022
2024
2026
year
c.
Th. $n = 1000$
mc. $n = 1000$
Th. $n = 10,000$
mc. $n = 10,000$
Th. $n = 100,000$
mc. $n = 100,000$
discovery rate
10^{-4}
10^{-29}
10^{-54}
10^{-7}
10^{-4}
10^{-1}
mutation rate μ
b.
Bigram conditional entropy
$H(X_\ell \mid X_{\ell-1})$ (bits/symbol)
mutation only
mutation+selection
2
1.5
1
0.5
0
200
400
generation
d.
discovery rate
Te^{-T}
10^{-4}
10^{-29}
10^{-54}
10^{-2}
10^{0}
10^{2}
mutation temperature $T = n\mu$
e.
$U = n\mu$
3
2
1
0
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
generation